

P1 - Proposed Solutions

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Problem 1.

Consider the local-linear estimator defined by

$$(\hat{m}_{LL}(x_0), \hat{m}'_{LL}(x_0)) = \arg \min_{\alpha, \beta} \sum_{i=1}^n K_h(X_i - x_0) [Y_i - \alpha - \beta(X_i - x_0)]^2,$$

where $K_h(u) = \frac{1}{h} K\left(\frac{u}{h}\right)$.

(a) Define

$$V \equiv \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad Z \equiv \begin{pmatrix} 1 & X_1 - x_0 \\ \vdots & \vdots \\ 1 & X_n - x_0 \end{pmatrix},$$

and

$$W \equiv \text{diag}\left(\frac{1}{K_h(X_1 - x_0)}, \dots, \frac{1}{K_h(X_n - x_0)}\right).$$

Then

$$\sum_{i=1}^n K_h(X_i - x_0) [Y_i - \alpha - \beta(X_i - x_0)]^2 = (V - Z\theta)' W^{-1} (V - Z\theta),$$

with $\theta = (\alpha, \beta)'$. The GLS minimizer is

$$\hat{\theta} = (Z' W^{-1} Z)^{-1} Z' W^{-1} V.$$

Since $\hat{\theta} = (\hat{m}_{LL}(x_0), \hat{m}'_{LL}(x_0))'$, we have shown that

$$\begin{pmatrix} \hat{m}_{LL}(x_0) \\ \hat{m}'_{LL}(x_0) \end{pmatrix} = (Z' W^{-1} Z)^{-1} Z' W^{-1} V.$$

(b) For fixed n , if $h \rightarrow \infty$, then

$$K_h(X_i - x_0) = \frac{1}{h} K\left(\frac{X_i - x_0}{h}\right) \rightarrow 0$$

because $K(0)$ is finite and the factor $1/h \rightarrow 0$. Since

$$W = \text{diag}(1/K_h(X_i - x_0)),$$

it follows that the diagonal elements of W diverge to $+\infty$, and hence $W^{-1} = \text{diag}(K_h(X_i - x_0)) \rightarrow 0$; that is, equal weights for all observations. More specifically, we know that a GLS with equal weights for each observation is a OLS!

(c) For the local-constant (Nadaraya-Watson) estimator we assume

$$h \rightarrow 0 \quad \text{and} \quad nh \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

The condition $h \rightarrow 0$ makes the estimator “local,” so the bias (which comes from averaging points farther from x_0) vanishes. The condition $nh \rightarrow \infty$ ensures that the effective number of observations inside the window of width h goes to infinity, which makes the variance go to zero. Thus one condition controls bias (localization) and the other controls variance (enough data in the neighborhood).

(d) Yes, the same two conditions $h \rightarrow 0$ and $nh \rightarrow \infty$ are also needed for the asymptotic theory of the local-linear estimator: we still need localization (to control bias) and we still need enough observations in the window (to control variance).

The main differences are: the local-linear estimator corrects for the first-order variation of $m(\cdot)$ around x_0 , so it has smaller bias—especially near the boundary—and it is less sensitive to the design (uneven density of X_i). The local-constant estimator is simpler but suffers from larger boundary/design bias; the local-linear one achieves the same variance order but with better bias properties.

Problem 2.

Consider the linear regression model

$$Y_i = X_i' \beta + u_i, \quad i = 1, \dots, n,$$

with OLS estimator $\hat{\beta}_n = (X'X)^{-1}X'Y$. We want inference on β .

(a) (Nonparametric/pairs bootstrap for $\sqrt{n}(\hat{\beta}_n - \beta)$)

1. The “units” we resample are the *pairs* (Y_i, X_i) , $i = 1, \dots, n$.
2. We resample *with replacement*: draw n indices $i_1, \dots, i_n$ from $\{1, \dots, n\}$ with replacement, and form a bootstrap sample $\{(Y_{i_\ell}, X_{i_\ell})\}_{\ell=1}^n$.
3. For this bootstrap sample, run OLS:

$$\hat{\beta}_n^{(b)} = (X^{(b)'} X^{(b)})^{-1} X^{(b)'} Y^{(b)},$$

where $X^{(b)}$ and $Y^{(b)}$ are the design matrix and outcome vector of the b -th resample.

4. Form the bootstrap analogue of the statistic:

$$R_n^{(b)} = \sqrt{n}(\hat{\beta}_n^{(b)} - \hat{\beta}_n).$$

5. Repeat for $b = 1, \dots, B$; the empirical distribution of $\{R_n^{(b)}\}_{b=1}^B$ estimates the distribution of $\sqrt{n}(\hat{\beta}_n - \beta)$.

(b) (Bootstrap t -test for $H_0 : \beta_j = 0$)

You computed, for each replication b ,

$$t_n^{(b)} = \frac{\hat{\beta}_{j,n}^{(b)} - \hat{\beta}_{j,n}}{\text{SE}^{(b)}(\hat{\beta}_{j,n}^{(b)})}.$$

Let t_n be the original (sample) t -statistic

$$t_n = \frac{\hat{\beta}_{j,n} - 0}{\text{SE}(\hat{\beta}_{j,n})}.$$

From the bootstrap $\{t_n^{(b)}\}_{b=1}^B$, compute the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles, call them

$$q_{\alpha/2} \quad \text{and} \quad q_{1-\alpha/2}.$$

A two-sided level- α test rejects H_0 if

$$t_n < q_{\alpha/2} \quad \text{or} \quad t_n > q_{1-\alpha/2}.$$

(c) (Bootstrap $(1 - \alpha)$ CI for β_j)

You have $\{R_n^{(b)}\}_{b=1}^B$ with

$$R_n^{(b)} = \sqrt{n}(\hat{\beta}_n^{(b)} - \hat{\beta}_n).$$

Let $c_{\alpha/2,j}$ and $c_{1-\alpha/2,j}$ be the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the j -th component of $\{R_n^{(b)}\}_{b=1}^B$. Since

$$\sqrt{n}(\hat{\beta}_{j,n} - \beta_j) \approx \text{bootstrap distribution},$$

we have, approximately,

$$c_{\alpha/2,j} \leq \sqrt{n}(\hat{\beta}_{j,n} - \beta_j) \leq c_{1-\alpha/2,j}.$$

Solving for β_j gives the $(1 - \alpha)$ confidence interval

$$\boxed{\hat{\beta}_{j,n} - \frac{c_{1-\alpha/2,j}}{\sqrt{n}} \leq \beta_j \leq \hat{\beta}_{j,n} - \frac{c_{\alpha/2,j}}{\sqrt{n}}}$$

which is the usual percentile- t style inversion for the pairs bootstrap in this setup.

Problem 3.

(a) Let

$$Y_t = X_t + e_t, \quad X_t = aX_{t-1} + u_t,$$

with $|a| < 1$, and where $\{u_t\}$ and $\{e_t\}$ are mutually independent i.i.d. sequences with

$$\mathbb{E}u_t = \mathbb{E}e_t = 0, \quad \text{Var}(u_t) = \sigma_u^2, \quad \text{Var}(e_t) = \sigma_e^2.$$

First write the MA(∞) representation of X_t :

$$X_t = \sum_{j=0}^{\infty} a^j u_{t-j},$$

so

$$Y_t = e_t + \sum_{j=0}^{\infty} a^j u_{t-j}.$$

Hence Y_t is a linear process whose autocovariances are easy to read:

- For $k \geq 1$:

$$\gamma_Y(k) = \text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(X_t, X_{t-k}) = a^k \frac{\sigma_u^2}{1-a^2},$$

because the e 's are independent over time and independent of X .

- For $k = 0$:

$$\gamma_Y(0) = \text{Var}(X_t) + \text{Var}(e_t) = \frac{\sigma_u^2}{1-a^2} + \sigma_e^2.$$

Now consider a generic ARMA(1,1) process with the *same* AR coefficient a :

$$Y_t = aY_{t-1} + v_t + \theta v_{t-1}, \quad v_t \sim \text{i.i.d. } (0, \sigma_v^2),$$

with $|\theta| < 1$. The autocovariances of an ARMA(1,1) with AR coefficient a are (standard formulas)

$$\gamma(0) = \frac{1 + \theta^2 + 2a\theta}{1 - a^2} \sigma_v^2, \quad \gamma(1) = \frac{(\theta + a)(1 + a\theta)}{1 - a^2} \sigma_v^2,$$

and for $k \geq 2$, $\gamma(k) = a^{k-1}\gamma(1)$.

To have the *same* autocovariance function as our $Y_t = X_t + e_t$, we must match

$$\gamma(1) = \frac{(\theta + a)(1 + a\theta)}{1 - a^2} \sigma_v^2 = \frac{a\sigma_u^2}{1 - a^2}$$

and

$$\gamma(0) = \frac{1 + \theta^2 + 2a\theta}{1 - a^2} \sigma_v^2 = \frac{\sigma_u^2}{1 - a^2} + \sigma_e^2.$$

From the first equality,

$$\sigma_v^2 = \frac{a\sigma_u^2}{(\theta + a)(1 + a\theta)}.$$

Plug this into the second equality to eliminate σ_v^2 ; after simplification you obtain the quadratic equation for θ :

$$a\sigma_e^2\theta^2 + (\sigma_u^2 + \sigma_e^2(1 + a^2))\theta + a\sigma_e^2 = 0.$$

Thus the MA coefficient must be

$$\theta = \frac{-(\sigma_u^2 + \sigma_e^2(1 + a^2)) \pm \sqrt{(\sigma_u^2 + \sigma_e^2(1 + a^2))^2 - 4a^2\sigma_e^4}}{2a\sigma_e^2}$$

and we choose the root with $|\theta| < 1$ (the other root is non-invertible). The AR coefficient is simply

$$\phi = a,$$

and the innovation variance is then

$$\sigma_v^2 = \frac{a\sigma_u^2}{(\theta + a)(1 + a\theta)}.$$

Therefore Y_t has the same autocovariance function as an ARMA(1,1) with AR coefficient a and MA coefficient given by the (invertible) root above.

(b) Consider

$$Y_t = -0.22 + 0.3Y_{t-1} + 0.5Y_{t-2} + 0.2Y_{t-3} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2).$$

(i) *Stationarity.* The AR polynomial is

$$1 - 0.3z - 0.5z^2 - 0.2z^3.$$

This polynomial factors as

$$1 - 0.3z - 0.5z^2 - 0.2z^3 = (1 - z)(1 + 0.7z + 0.2z^2).$$

Hence it has a root at $z = 1$, i.e. a unit root. Because of this unit root, the process $\{Y_t\}$ is *not stationary*, so we cannot write its finite, time-invariant autocovariance function.

(ii) *Dynamics of $\Delta Y_t = Y_t - Y_{t-1}$.* Since

$$(1 - z)(1 + 0.7z + 0.2z^2)Y_t = -0.22 + \varepsilon_t,$$

and $(1 - z)Y_t = \Delta Y_t$, we get

$$(1 + 0.7L + 0.2L^2)\Delta Y_t = -0.22 + \varepsilon_t,$$

or, in time domain,

$$\Delta Y_t + 0.7\Delta Y_{t-1} + 0.2\Delta Y_{t-2} = -0.22 + \varepsilon_t,$$

i.e.

$$\Delta Y_t = -0.7\Delta Y_{t-1} - 0.2\Delta Y_{t-2} - 0.22 + \varepsilon_t.$$

This is an AR(2) with constant and white-noise innovations. The roots of

$$1 + 0.7z + 0.2z^2 = 0$$

are the two non-unit roots of the original cubic, and they lie outside the unit circle, so this AR(2) is *stationary*. Thus ΔY_t is stationary AR(2).

Problem 4.

Consider

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots,$$

where $\varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$, $\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \cdots$ is invertible, $\sum_{i=1}^{\infty} |\theta_i| < \infty$, and we observe $\{Y_1, \dots, Y_T\}$, with $\varepsilon_t = 0$ for $t \leq 0$ and $Y_0 = \mu$.

(a) Let $\mathcal{F}_t = \sigma(Y_s : s \leq t)$ and $\mathcal{E}_t = \sigma(\varepsilon_s : s \leq t)$. Because $\theta(L)$ is invertible, we can write

$$\varepsilon_t = \theta(L)^{-1}(Y_t - \mu)$$

as a convergent linear function of $\{Y_t, Y_{t-1}, \dots\}$. Hence knowing $\{Y_s : s \leq t\}$ is equivalent to knowing $\{\varepsilon_s : s \leq t\}$, and conversely $Y_t = \mu + \theta(L)\varepsilon_t$ is a linear function of present and past shocks. Therefore

$$\boxed{\mathcal{F}_t = \mathcal{E}_t \quad \text{for all } t.}$$

(b) For $h \geq 1$,

$$Y_{t+h} = \mu + \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \theta_2 \varepsilon_{t+h-2} + \cdots.$$

Given $\mathcal{F}_t = \mathcal{E}_t$, at time t we know all $\varepsilon_t, \varepsilon_{t-1}, \dots$, but we do *not* know $\varepsilon_{t+1}, \dots, \varepsilon_{t+h}$. Hence

$$\hat{Y}_{t+h|t} \equiv \mathbb{E}[Y_{t+h} | \mathcal{F}_t] = \mu + \theta_h \varepsilon_t + \theta_{h+1} \varepsilon_{t-1} + \theta_{h+2} \varepsilon_{t-2} + \cdots,$$

i.e. we keep only the terms whose shocks are dated $\leq t$.

The forecast error is

$$e_{t+h|t} \equiv Y_{t+h} - \hat{Y}_{t+h|t} = \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \theta_2 \varepsilon_{t+h-2} + \cdots + \theta_{h-1} \varepsilon_{t+1}.$$

This is a *finite* MA process of order $h-1$ with coefficients $(1, \theta_1, \dots, \theta_{h-1})$:

$$\boxed{e_{t+h|t} = \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \cdots + \theta_{h-1} \varepsilon_{t+1}.}$$

A finite MA is stationary, so $\{e_{t+h|t}\}$ is stationary. Its autocovariances clearly depend on $\theta_1, \dots, \theta_{h-1}$, so the autocovariance structure *does* depend on the MA parameters.

(c) Consider the forecasting regression

$$Y_{t+h} = \beta_0 + \beta_1 X_t + u_{t+h}, \quad \mathbb{E}[u_{t+h} | X_t] = 0,$$

estimated for $t = 1, \dots, T-h$. Even if X_t is exogenous, the regression error u_{t+h} will contain the same innovation part as the optimal forecast error in (b), namely

$$u_{t+h} = \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \cdots + \theta_{h-1} \varepsilon_{t+1} + (\text{possibly an } X_t\text{-orthogonal part}).$$

Therefore, across t the errors overlap in shocks. For example,

$$u_{t+h} = \varepsilon_{t+h} + \theta_1 \varepsilon_{t+h-1} + \cdots + \theta_{h-1} \varepsilon_{t+1},$$

$$u_{t+1+h} = \varepsilon_{t+1+h} + \theta_1 \varepsilon_{t+h} + \cdots + \theta_{h-1} \varepsilon_{t+2},$$

so u_{t+h} and u_{t+1+h} share the shocks $\varepsilon_{t+h}, \varepsilon_{t+h-1}, \dots, \varepsilon_{t+2}$. Hence

$\{u_{t+h}\}$ follows an $\text{MA}(h-1)$ process (serial correlation up to lag $h-1$).

(d) OLS standard errors are valid only under no serial correlation (and homoskedasticity). Here the regression errors are $\text{MA}(h-1)$, so the usual OLS formula will *underestimate* the true variance of $\hat{\beta}_1$. We therefore need heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors.

Because we know the error is $\text{MA}(h-1)$, it is natural to use a Newey–West/HAC estimator with truncation lag

$$L \geq h - 1.$$

This ensures that all nonzero autocovariances of the error are accounted for. Thus:

Use HAC (e.g. Newey–West) standard errors with bandwidth at least $h-1$.

Problem 5.

Consider

$$Y_t = X_t' \beta + u_t, \quad t = 1, \dots, T,$$

with $\mathbb{E}[u_t X_t] = 0$ and X_t is $k \times 1$.

(a) Define the sample moment

$$g_T(\beta) = \frac{1}{T} \sum_{t=1}^T X_t (Y_t - X_t' \beta) = \frac{1}{T} \sum_{t=1}^T X_t Y_t - \left(\frac{1}{T} \sum_{t=1}^T X_t X_t' \right) \beta.$$

Let

$$\bar{q}_T \equiv \frac{1}{T} \sum_{t=1}^T X_t Y_t, \quad S_{XX} \equiv \frac{1}{T} \sum_{t=1}^T X_t X_t'.$$

Then $g_T(\beta) = \bar{q}_T - S_{XX} \beta$.

Given a symmetric positive definite weighting matrix W_T , the GMM objective is

$$J_T(\beta) = g_T(\beta)' W_T g_T(\beta).$$

The first-order condition is

$$\frac{\partial J_T(\beta)}{\partial \beta} = -2 S_{XX}' W_T g_T(\beta) = 0 \implies S_{XX}' W_T (\bar{q}_T - S_{XX} \beta) = 0.$$

Since S_{XX} is symmetric, $S_{XX}' = S_{XX}$, so

$$S_{XX} W_T S_{XX} \beta = S_{XX} W_T \bar{q}_T,$$

and therefore

$$\boxed{\hat{\beta}_{\text{GMM}} = (S_{XX} W_T S_{XX})^{-1} S_{XX} W_T \bar{q}_T}.$$

(b) If u_t are i.i.d. with $\mathbb{E}[u_t^2 | X_t] = \sigma^2$, then

$$\Omega \equiv \text{Var}(\sqrt{T} g_T(\beta_0)) = \mathbb{E}[u_t^2 X_t X_t'] = \sigma^2 \mathbb{E}[X_t X_t'].$$

Hence the optimal weighting matrix is

$$W^* = \Omega^{-1} = (\sigma^2 \mathbb{E}[X_t X_t'])^{-1} = \frac{1}{\sigma^2} (\mathbb{E}[X_t X_t'])^{-1}.$$

Plugging W^* into the sample formula (replacing expectations by sample averages) gives

$$\hat{\beta}_{\text{GMM}} = (S_{XX} W^* S_{XX})^{-1} S_{XX} W^* \bar{q}_T = (S_{XX} (\frac{1}{\sigma^2} S_{XX}^{-1}) S_{XX})^{-1} S_{XX} (\frac{1}{\sigma^2} S_{XX}^{-1}) \bar{q}_T = S_{XX}^{-1} \bar{q}_T.$$

But

$$S_{XX}^{-1} \bar{q}_T = \left(\frac{1}{T} \sum_{t=1}^T X_t X_t' \right)^{-1} \left(\frac{1}{T} \sum_{t=1}^T X_t Y_t \right) = \left(\sum_{t=1}^T X_t X_t' \right)^{-1} \sum_{t=1}^T X_t Y_t,$$

which is the OLS estimator. Thus, with i.i.d. homoskedastic errors, optimal GMM = OLS.

(c) If the errors are heteroskedastic but not serially correlated, i.e.

$$\mathbb{E}[u_t^2 | X_t] = \sigma_t^2 \quad \text{and} \quad \mathbb{E}[u_t u_s | X_t, X_s] = 0 \text{ for } t \neq s,$$

then

$$\Omega = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sigma_t^2 X_t X_t',$$

so the optimal weighting matrix is

$$W^* = \Omega^{-1}.$$

In practice, we estimate β first (e.g. by OLS), get residuals $\hat{u}_t = Y_t - X_t' \hat{\beta}$, and form the heteroskedasticity-robust estimate

$$\hat{\Omega} = \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 X_t X_t'.$$

Then set

$$\hat{W} = \hat{\Omega}^{-1}$$

and plug this $\hat{W}$ into the GMM formula to get the efficient (two-step) GMM estimator. This is the usual White/Huber-Eicker type correction.

(d) If the errors are both heteroskedastic and serially correlated (with correlation decaying in the lag), then

$$\Omega = \sum_{\ell=-\infty}^{\infty} \Gamma_{\ell}, \quad \Gamma_{\ell} = \mathbb{E}[u_t u_{t-\ell} X_t X_{t-\ell}'].$$

That is, Ω is the long-run covariance matrix of $X_t u_t$. The optimal weighting matrix is again

$$W^* = \Omega^{-1}.$$

To estimate Ω in practice, use a HAC (Newey-West-type) estimator. With residuals $\hat{u}_t$,

$$\hat{\Gamma}_0 = \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 X_t X_t', \quad \hat{\Gamma}_{\ell} = \frac{1}{T} \sum_{t=\ell+1}^T \hat{u}_t \hat{u}_{t-\ell} X_t X_{t-\ell}' \quad (\ell = 1, 2, \dots),$$

and choose a kernel $k(\cdot)$ (e.g. Bartlett) and a bandwidth L ; then

$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{\ell=1}^L k\left(\frac{\ell}{L+1}\right) (\hat{\Gamma}_{\ell} + \hat{\Gamma}_{\ell}').$$

Finally,

$$\hat{W} = \hat{\Omega}^{-1},$$

and this $\hat{W}$ is used in the (two-step) GMM estimator. The main choices are: (i) the HAC kernel, and (ii) the truncation lag/bandwidth L (often taken to grow slowly with T).